

André

Fonseca Cavalcanti

Mechanical Engineer · Software Dev.
Industrial Manager

CONTACT

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EDUCATION

- 2023 – 2025
Professional M.Sc.
Science, Technology & Innovation · UFRN
- 2018 – 2021
B.Eng. Mechanical Engineering
UFRN

- 2012 – 2017
B.Sc. Science and Technology
UFRN

LANGUAGES

- Portuguese Native
- English Fluent

SKILLS

ENGINEERING / CAD

- Fusion 360
- SolidWorks
- Creo Parametric
- Scilab

DEVELOPMENT

- Python
- TypeScript
- React
- FastAPI
- Node.js
- SQL
- Firebase
- Next.js

DATA & AI

- ANN / SOM
- Prophet
- XGBoost
- Metabase
- Excel / VBA

MANAGEMENT

- Lean
- PDCA
- MPC
- CMMS
- Industry 4.0

AWARDS

- 2013
Champion — IEEE Very Small Size
Brazilian Robotics Competition

EXPERIENCE

Soluções Corporativas Integradas LTDA

Oct 2025 – Present

Partner · Operations Director

Partner in a technology company for industrial and business management software, responsible for strategy and technical roadmap. Leads four SaaS products: ManOS (industrial CMMS), SOFIN (AI-powered financial management), CompraCerta (retail procurement) and OpenInsurance (insurance management).

Nordeste Fruit LTDA

May 2025 – Oct 2025

Industrial Manager

Eliminated stockouts and reduced raw material losses by restructuring demand and production control processes. Leading a team of 30 employees, improved factory energy efficiency and operational costs.

Implasverde Indústria de Plásticos Baixa Verde LTDA

May 2022 – May 2025

Industrial Manager

Led a multifunctional team of 160 employees across production and support departments. The application of Lean and PDCA methodologies promoted continuous improvement in quality and efficiency indicators.

Implasverde Indústria de Plásticos Baixa Verde LTDA

May 2021 – May 2022

Maintenance Supervisor

Led the implementation of MPC (Maintenance Planning and Control) with a 10-person team, improving equipment availability and reliability across the plant.

Guararapes Confeções S.A.

May 2019 – Apr 2020

Engineering Intern

Developed an integrated solution to automate production control in the cutting department, eliminating manual reporting and creating real-time KPI dashboards to support strategic decision-making.

Instituto Metr pole Digital – UFRN

Feb 2018 – Feb 2021

Research Fellow · Prototyping Laboratory

Coordinated the NEDGlove project – a computational glove for virtual motor rehabilitation, integrating sensor hardware with gamification software for physiotherapy. Worked on electronic and mechanical prototype development for university research.

Barreira do Inferno Launch Center

Aug 2015 – Jan 2016

Research Intern · DCTA / Brazilian Air Force

Developed, within the SPACEVANT project, an automatic cargo launch system for unmanned aerial vehicles, enhancing DCTA's aerospace mission capabilities. The work was published at CONEM 2016.

ACADEMIC PROFILE

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RESEARCH AREAS

Industry 4.0 & Autonomous Maintenance

Cyber-Physical Systems

Applied Artificial Intelligence

Process Optimisation

Autonomous Mobile Robotics

FIELDS OF WORK

Mechanical Engineering

Computer Science

Business Administration

Industrial Economics

PUBLICATIONS

7 papers at national and international conferences: robotics, AI, industrial maintenance and rehabilitation.

5 citations (Scholar, Jan/2014)

COMPANY

Soluções Corporativas

Integradas LTDA

Partner · Operations Director

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SOFTWARE PROJECTS

ManOS — Industrial CMMS

Web-based maintenance management with 8 dashboards, MTTR/MTBF/RPN calculation and ISO 14224 & ISO 55000 compliance. Demo with 50+ assets and 1,000+ work orders.

React TypeScript Firebase Nodejs

CompraCerta — Procurement Intelligence

B2B SaaS for supermarkets with demand forecasting via Prophet + XGBoost ensemble, supplier scoring and purchase orders with multi-tenancy.

FastAPI Python PostgreSQL React

SOFIN — Financial Management

SaaS for SMEs with generative AI (Gemini), invoice OCR, automatic bank reconciliation, P&L statements and client-side AES-GCM encryption.

React TypeScript Firebase Gemini AI

OpenInsurance — Insurance Management

Insurance quotation and policy management platform with Zod schema validation and Prisma ORM.

Nextjs TypeScript Prisma

RESEARCH & INNOVATION

Machinery Guardian · Plastics manufacturer, RN

2023 – 2025

Autonomous Maintenance using SOM Neural Networks

Eight months on a bag-making machine, ERP-integrated: breakdowns from 18 to about 3 per month, spare-parts cost down 50%, inspection 13 min vs. 248 min of downtime per breakdown.

Research project · ABS Wind do Brasil

2021 – 2023

Wind turbine gearboxes, SOM · Coord.: Prof. E. P. Matamoras

Heart Disease Detection with Neural Networks

2020 – 2022

Member · UFRN

Member of the team of five undergraduates building an unsupervised SOM heart disease classifier; coordinator: Prof. M. A. L. Cabral.

PUBLICATIONS

[1] A. Cavalcanti, V. Landim, J. Medeiros Neto, J. Dantas, and G. L. A. d. Albuquerque. Cargo transport and launch support from an unmanned aerial vehicle. In *IX Brazilian National Congress of Mechanical Engineering (CONEM)*, Fortaleza, Brazil, 2016.

[2] A. F. Cavalcanti. Unsupervised autonomous maintenance system using artificial intelligence. In *Undergraduate Programme in Mechanical Engineering – UFRN*, 2021. Final Year Project.

[3] A. F. Cavalcanti. Machinery Guardian: Optimising maintenance with artificial intelligence and autonomy. In *Professional Graduate Programme in Science, Technology and Innovation – UFRN*, 2025. M.Sc. Dissertation.

[4] A. F. Cavalcanti, M. V. Araújo, and D. P. F. Pedrosa. Small wheeled autonomous mobile soccer robot. In *Joint Conference on Robotics: SBR-LARS Robotics Symposium and Robo-control*, page 49, São Carlos, Brazil, 2014.

[5] A. F. Cavalcanti, F. B. S. d. Medeiros, and R. R. Dantas. Evaluate Leap Motion control for multiple hand posture recognition. In *19th Symposium on Virtual and Augmented Reality (SVR)*, Curitiba, Brazil, 2017.

[6] A. F. Cavalcanti, A. S. d. Souza, A. P. V. Pinto, L. P. W. d. Oliveira, R. B. Julião, M. A. L. Cabral, E. P. Matamoras, and E. A. R. Seabra. Detection and classification of water chiller failures using artificial neural networks. In *7th International Conference on Integrity-Reliability-Failure (IRF2020)*, pages 91–92, Funchal, Portugal, 2020.

[7] L. F. d. A. Santos, J. F. Rangel, A. F. Cavalcanti, K. L. d. Bessa, and W. B. d. A. Santos. Average superficial roughness analysis of manufactured pipes by 3D printing. In *25th ABCM International Congress of Mechanical Engineering (COBEM 2019)*, Uberlândia, Brazil, 2019.